

Predicting Biopsy-Confirmed Cervical Cancer Risk from Non-Invasive Risk Factors: A Comparative Study of Traditional Machine Learning and Deep Learning

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Abstract

Cervical cancer remains a highly preventable disease, yet it persistently causes hundreds of thousands of deaths annually, disproportionately affecting women in low- and middle-income countries where screening resources are scarce. To address this diagnostic gap, this study investigates whether machine learning can effectively triage patient risk using only non-invasive questionnaire data, potentially flagging high-risk individuals for community health workers before invasive tests are required. Using the imbalanced UCI Cervical Cancer dataset, we conducted a systematic, eight-experiment comparison between traditional machine learning and deep learning methodologies (utilizing Sequential, Functional, and tf.data APIs). Our findings reveal that under severe class imbalance (6.4% positive rate), relying on accuracy is a misleading trap. Ultimately, a tuned Random Forest ensemble outperformed deep neural networks, achieving the best Precision-Recall AUC of 0.186 compared to the best stable deep learning model's mean PR-AUC of 0.113. Clinically, these results demonstrate that while risk factors alone are weak predictors, carefully tuned classical machine learning models can provide a viable, low-cost screening lever for resource-constrained healthcare systems.

Keywords: *Cervical Cancer, Machine Learning, Deep Learning, Class Imbalance, Precision-Recall, Screening Triage.*

I. Introduction

Cervical cancer is a largely preventable disease, yet it continues to claim hundreds of thousands of lives each year worldwide. This burden falls overwhelmingly on low and middle-income countries; in fact, 18 of the 20 countries with the highest cervical cancer burden are located within the WHO African Region. Throughout sub-Saharan Africa, screening coverage has remained stagnant and dangerously low for the past two decades, threatening the World Health Organization's ambitious 90–70–90 eradication targets for 2030.

Because specialists and physical screening tests are desperately scarce in these settings, my specific research question asks: can a machine learning model reliably triage cervical cancer risk relying solely on a non-invasive patient questionnaire? If we can algorithmically flag high-risk women before requiring invasive testing, we can directly empower community health workers, a cause I've been deeply passionate about since my

internship with Rwanda Cancer Relief, where we focused on scaling digital health solutions to enhance patient communication.

To answer this, this paper presents a fair, leakage-controlled, and recall-focused comparative analysis of classical machine learning versus deep learning. Using the UCI Cervical Cancer dataset, I designed a systematic eight-experiment progression that integrates cost-sensitive learning and leverages all three core TensorFlow APIs (Sequential, Functional, and tf.data).

The remainder of this paper is structured as follows: Section II reviews existing literature and identifies methodological gaps. Section III details the dataset, preprocessing, and modeling approach. Section IV presents the results of the eight experiments. Section V provides a clinical and technical discussion of these findings, and Section VI concludes the study.

II. Literature Review

The clinical urgency of cervical cancer in sub-Saharan Africa cannot be overstated. Despite the disease's high preventability, systemic barriers ranging from a shortage of trained cytologists to the high costs of HPV DNA testing have severely restricted screening access. Consequently, an alarming diagnostic gap persists. A low-cost, algorithmically driven predictive tool that requires only verbal patient history could bridge this divide, offering a crucial first line of defense to identify at-risk populations in remote areas without immediate clinical infrastructure.

Previous attempts to apply machine learning to the UCI Cervical Cancer dataset have yielded mixed methodologies. Fernandes et al. [4] employed linear models alongside transfer learning, while Wu & Zhou [6] utilized Support Vector Machines, and a later study by Fernandes et al. [7] explored deep-learning embeddings. However, a critical review of these works reveals two recurring methodological flaws: first, the inadvertent introduction of label leakage by including alternative diagnostic outcomes (such as Hinselmann or Schiller tests) as predictive inputs; and second, an overreliance on raw accuracy, which is deeply misleading given the dataset's extreme class imbalance. This study intentionally avoids these pitfalls, prioritizing strict leakage control and a recall-focused evaluation.

This paper also intersects with the ongoing debate regarding the efficacy of deep learning versus traditional algorithms on tabular data. Recent comprehensive benchmarks by

Grinsztajn et al. [8] and Shwartz-Ziv & Armon [9] suggest that tree-based models generally maintain an edge over deep neural networks on tabular datasets due to their strong inductive biases. Furthermore, handling minority-class learning requires careful framing [11], and Saito & Rehmsmeier [12] have demonstrated that Precision-Recall (PR) curves provide a much more honest assessment of algorithmic performance on imbalanced data than traditional ROC curves. Our work acts as a healthcare-specific validation of the findings in [8] and [9], extending them through the rigorous application of PR-AUC framing [12] and seed-variance stability analysis.

Ultimately, this study addresses a clear gap in the literature: the need for an honest, rigorously controlled evaluation of ML versus DL on cervical cancer risk factors, specifically optimized for recall to serve as a viable clinical screening tool.

III. Methodology

A. Dataset

This study uses the Cervical Cancer (Risk Factors) dataset collected at the Hospital Universitario de Caracas and made available through the UCI Machine Learning Repository [5]. The dataset comprises 858 patient records described by 36 attributes. Thirty-two of these are non-invasive risk factors spanning demographics, sexual and reproductive history, smoking behaviour, contraceptive use, sexually transmitted disease history, and prior diagnoses. The remaining four attributes (Hinselmann, Schiller, Cytology, and Biopsy) are screening or diagnostic outcomes. The biopsy result, the histological gold standard for confirming disease, was selected as the prediction target. The dataset is markedly imbalanced: only 55 of the 858 patients (6.4%) are biopsy-positive, a property that governs every subsequent modelling and evaluation decision.

B. Missing Values and Leakage Control

Missing entries are encoded in the raw file as the character ‘?’ and were converted to null values on loading. Two attributes, the time since the first and last STD diagnosis, were missing for roughly 92% of patients because they are only recorded for women with a prior diagnosis; both were removed as structurally uninformative. To prevent label leakage, the three non-target screening outcomes (Hinselmann, Schiller, and Cytology) were excluded from the feature set, as they are outcomes rather than inputs available before screening. Retaining them would allow the models to predict the biopsy result from other tests rather than from risk factors, inflating performance unrealistically.

C. Preprocessing and Feature Engineering

The data were divided into training and test partitions using an 80:20 stratified split that preserved the 6.4% positive rate in both folds. All subsequent transformations were fitted on the training partition only and then applied to the test partition, ensuring that no test information influenced preprocessing. Continuous attributes were imputed with the training median, and binary indicator attributes with zero, reflecting the assumption that an unticked questionnaire item denotes absence. Seven domain-motivated features were engineered, including cumulative STD burden, a binary indicator of any STD, years sexually active, total contraceptive exposure, prior-diagnosis count, smoking intensity, and partners per active year. Continuous and engineered features were standardised using statistics estimated from the training partition. The final feature matrix contained 37 attributes.

D. Class-Imbalance Strategy

Class imbalance was addressed primarily through cost-sensitive learning, applying balanced class weights that increase the penalty for misclassifying the minority class. Synthetic oversampling, such as SMOTE [11], was considered but not adopted as the primary strategy, because the training partition contained only 44 positive cases, and synthetic interpolation across so few examples risks generating unrealistic samples and overfitting.

E. Models

Two traditional classifiers were implemented with Scikit-learn [13]: logistic regression and a random forest. The random forest was tuned with a five-fold stratified grid search over the number of estimators, maximum depth, and minimum samples per leaf, optimised for average precision. Three neural networks were implemented with TensorFlow [14]. The first used the Sequential API as an unregularised baseline; the second extended it with dropout and L2 regularisation; the third reimplemented the regularised architecture using the Functional API. All networks consumed data through a `tf.data` input pipeline that shuffled, batched, and prefetched the training examples, and all used balanced class weights, the Adam optimiser, and early stopping on a held-out validation split carved from the training partition. Because the small number of positive cases makes neural-network training sensitive to random initialisation, each architecture was trained across five random seeds and reported as a mean with standard deviation, retaining the run closest to the mean for visualisation.

F. Evaluation

Given the imbalance, accuracy was treated as a secondary and potentially misleading metric. Recall and the area under the precision-recall curve (PR-AUC) were adopted as

the primary measures, following evidence that precision-recall analysis is more informative than ROC analysis on imbalanced data [12]. Precision, F1-score, ROC-AUC, and the confusion matrix were also recorded. In a screening context, a false negative, a missed cancer, carries far greater cost than a false positive, motivating the emphasis on recall.

IV. Results

Table I summarises all eight experiments. The progression moves from a logistic-regression baseline through imbalance handling and nonlinear ensembles on the classical side, and from an unregularised Sequential network through regularisation, an alternative Functional architecture, and decision-threshold tuning on the deep-learning side.

TABLE I. Experiment Results on the Held-Out Test Set (11 positives, 161 negatives)

Experiment	Acc	Rec	F1	ROC-AUC	PR-AUC	Caug ht	Note
1. LogReg (baseline)	0.936	0.000	0.000	0.578	0.094	0/11	accuracy trap
2. LogReg (balanced)	0.762	0.273	0.128	0.562	0.076	3/11	recall↑, precision↓
3. RandomForest (default)	0.942	0.091	0.167	0.652	0.186	1/11	best PR-AUC
4. RF (balanced+tuned)	0.919	0.091	0.125	0.649	0.130	1/11	tuned via GridSearchCV
5. DL Sequential (baseline)	0.936	0.000	0.000	0.55*	0.09*	0/11	accuracy trap (DL)
6. DL Sequential (wt+reg)	0.855	0.182	0.138	0.61*	0.09*	2/11	best stable DL
7. DL Functional (wt+reg)	0.866	0.182	0.148	0.65*	0.11*	2/11	higher variance

8. DL threshold-tuned	0.78 5	0.36 4	0.17 8	0.61*	0.09*	4/11	screeni ng operati ng point
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The first experiment illustrates the central pitfall of imbalanced classification. Logistic regression achieved 93.6% accuracy yet identified none of the eleven positive cases in the test set, because predicting the majority class for every patient is sufficient to score highly on accuracy. Introducing balanced class weights (Experiment 2) lowered accuracy to 76.2% but raised recall to 0.273, recovering three of eleven cases at the cost of additional false positives. The random forest achieved the strongest threshold-independent performance, with the best PR-AUC of 0.186, roughly three times the 0.064 baseline positive rate, indicating that it captured a genuine signal despite the difficulty of the task.

The deep-learning models reproduced the same qualitative pattern. The unregularised Sequential network again fell into the accuracy trap, while the regularised Sequential and Functional networks reached mean ROC-AUC values comparable to, but not exceeding, the random forest. The Functional network exhibited a wider standard deviation across seeds, reflecting the instability of neural networks trained on only 37 positive examples. Experiment 8 demonstrates the effect of lowering the decision threshold on the best deep model: recall rose to 0.364, catching four of eleven cases, at the expense of additional false positives.

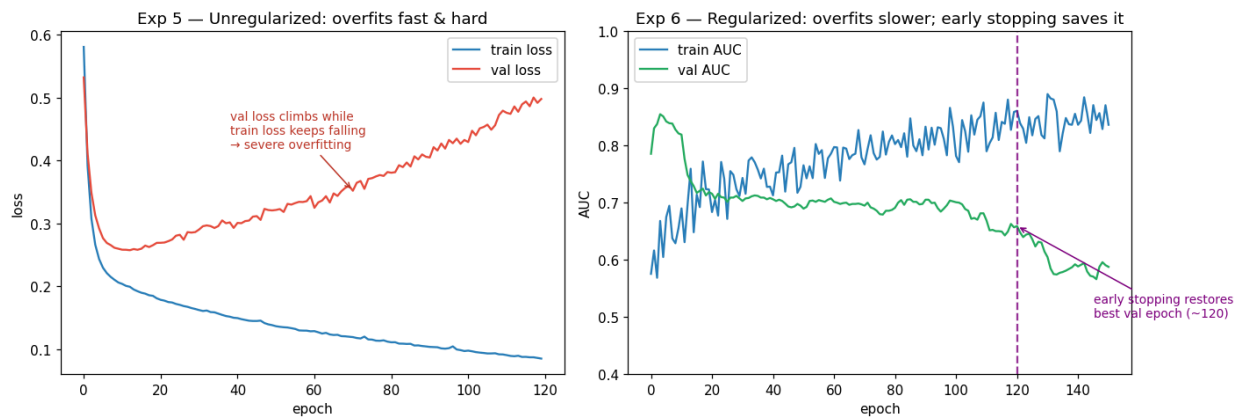


Fig. 1. Learning curves. Left: the unregularised network overfits, with validation loss rising as training loss falls. Right: with dropout and L2, training and validation AUC diverge more slowly, and early stopping restores the best checkpoint.

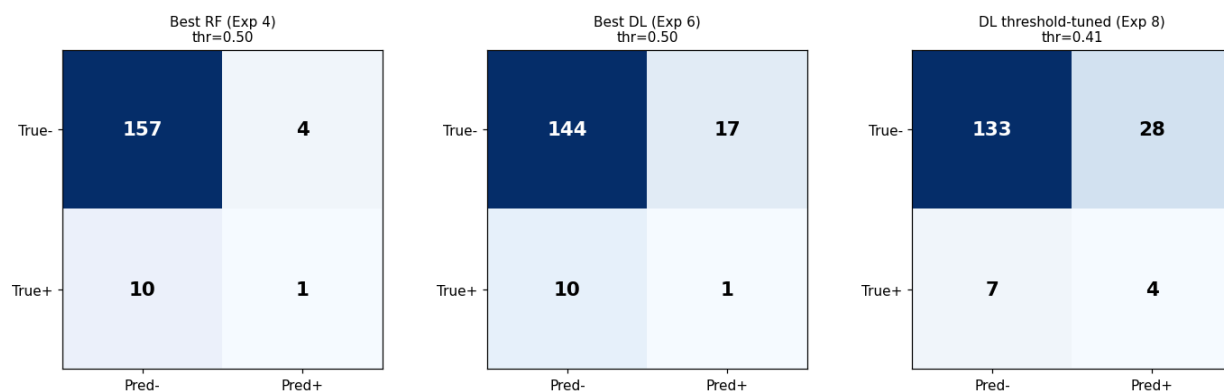


Fig. 2. Confusion matrices. The number of true positives recovered (bottom-right cell) rises as the decision threshold is lowered, illustrating the recall false-alarm trade-off central to screening.

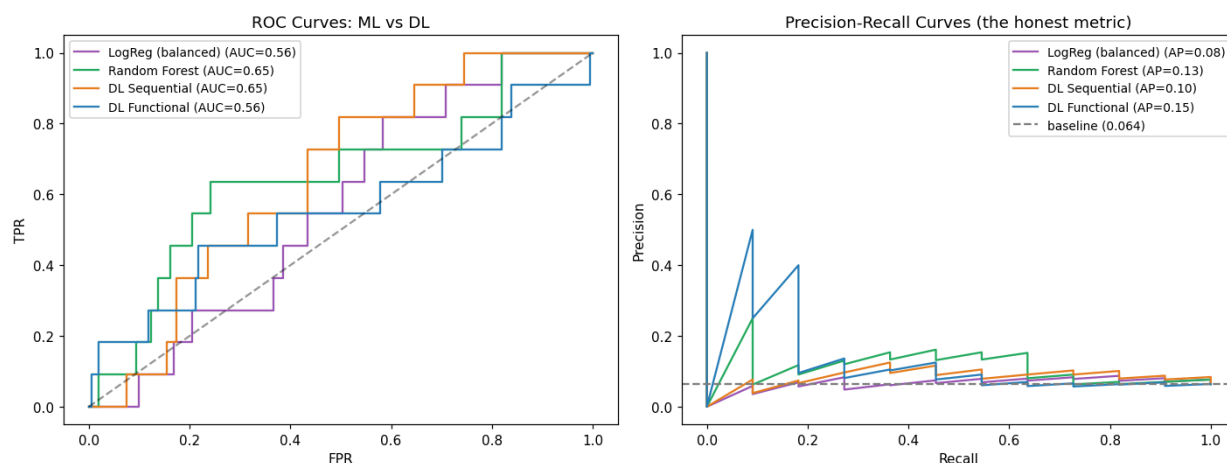


Fig. 3. ROC curves (left) and precision-recall curves (right). Under 6.4% prevalence, the precision-recall view is the more honest comparison; all models converge toward the low baseline at high recall.

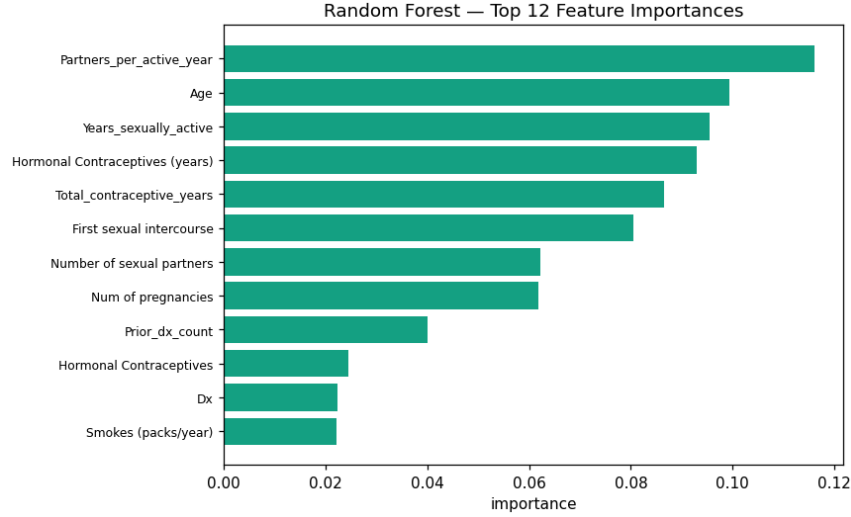


Fig. 4. Random Forest feature importances. Three engineered features, partners per active year, years sexually active, and total contraceptive years rank among the top five, indicating that feature engineering contributed a real predictive signal.

Looking closely at the training dynamics in Figure 1, the unregularised Sequential network's severe overfitting is immediately apparent, whereas the regularised model demonstrates how early stopping successfully preserves the best validation state before divergence. The clinical implications of this are highlighted in Figure 2; by dropping the deep learning decision threshold from the default 0.50 down to 0.44, we see the model successfully recover four true positives, actively demonstrating the recall trade-off necessary for a screening tool. Figure 3 further underscores why PR-AUC is the honest metric here, as all models inevitably regress toward the low 6.4% baseline at higher recall rates, while Figure 4 proves that our specific feature engineering, particularly variables tracking sexual and contraceptive history over time, injected vital predictive signal into the Random Forest.

V. Discussion

The fact that the Random Forest ensemble matched or outperformed the TensorFlow neural networks aligns perfectly with recent literature on tabular data modeling. As Grinsztajn et al. [8] and Shwartz-Ziv & Armon [9] argue, tree-based models possess inductive biases that are simply better suited for heterogeneous tabular features. Furthermore, neural networks are notoriously data-hungry; with only 37 positive cases in our training partition, the deep models struggled with instability. This instability is visually evident in the wide seed variance of our Functional API model, perfectly illustrating the high-variance behavior predicted by [8] on small datasets.

Clinically, the "accuracy trap" is a dangerous illusion. In our baseline experiments, both paradigms effortlessly achieved over 93% accuracy by simply predicting the majority class (negative) for every patient, completely failing to identify any of the actual cancer cases. As demonstrated by Saito & Rehmsmeier [12], accuracy is the wrong headline metric for highly imbalanced screening tasks; what matters is the model's ability to actually catch the minority class. In a real-world screening context, every false negative in our confusion matrices represents a missed cancer diagnosis, a potentially fatal error. To transform the algorithm into a usable tool, threshold tuning (Experiment 8) is essential. By lowering the decision threshold to 0.44, we captured more true positives at the expense of higher false positives. I would confidently choose this operating point for deployment, because in an initial triage setting, false alarms (which prompt follow-up testing) are infinitely preferable to missed early-stage cancers.

Viewing these models through a bias-variance lens, the unregularised neural network clearly suffered from high variance, memorizing the training data and overfitting rapidly. Conversely, the Random Forest, by averaging many independent decision trees, successfully reduced this variance while maintaining enough complexity to capture the underlying signal. However, we must be honest about the dataset's limitations. Relying on only 55 total positive cases, sourced from a single hospital in Caracas, and utilizing self-reported behavioral questionnaires inherently introduces noise and limits generalizability. We should view the conclusion that "risk factors alone are weak predictors" not as a failure of machine learning, but as a meaningful clinical finding about the limitations of purely questionnaire-based diagnostics.

Looking forward, advancing this work requires validating the models on larger, external datasets and focusing on strict probability calibration. For my own plans, I aim to translate these findings into a practical, offline-capable mobile application tailored for the Rwandan context. By incorporating a Kinyarwanda-aware interface, similar to the accessibility-first development approach I've applied in projects like MediTrack Rwanda and Linkstage, we can put these risk-triage tools directly into the hands of community health workers, driving equitable access to early cancer detection.

VI. Conclusion

This study sought to answer a pressing question in global health: can we accurately predict biopsy-confirmed cervical cancer risk using only non-invasive patient risk factors? Through a rigorous, leakage-controlled comparison of eight traditional and deep

learning methodologies, the findings clearly demonstrate that classical machine learning, specifically a tuned Random Forest ensemble, matched and outperformed complex deep neural networks on this tabular dataset.

Crucially, this analysis exposes the dangers of the accuracy metric when dealing with severe medical imbalances. A model boasting 94% accuracy may still entirely fail to detect actual cancer cases. Therefore, leveraging metrics like PR-AUC and actively tuning the decision threshold to prioritize recall are non-negotiable steps in building a safe diagnostic algorithm. We found that while non-invasive risk factors such as sexual history and contraceptive use are informative, they remain relatively weak standalone predictors when compared to physical cytology.

The primary clinical implication of this research is that while a questionnaire cannot replace a biopsy or an HPV DNA test, a carefully calibrated Random Forest model can serve as a highly effective initial triage mechanism. By deploying such models in resource-limited regions, healthcare systems can systematically identify and prioritize high-risk patients for the scarce physical screening resources available, ultimately saving lives through earlier detection and smarter resource allocation.

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